

Vibrational constants of AlO from its electronic band spectrum

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 Least count of the comparator **0.001 cm**

Aim

To determine the vibrational constants of the aluminium oxide molecule for the upper ($B^2\Sigma^+$) and lower ($X^2\Sigma^+$) electronic states from the band heads of its visible band system.

Observations



The mercury comparison spectrum, with the lines measured marked and labelled by wavelength in Å.

Table 1 - comparator readings of the mercury standard lines

Line	λ (Å)	M.S.R. (cm)	V.S.D.	d (cm)
Hg Violet	4046.56	2.85	0	2.850
Hg Violet (weak)	4077.83	3.35	0	3.350
Hg Blue	4358.33	7.00	47	7.047
Hg Blue-green (weak)	4916.04	11.75	40	11.790
Hg Green	5460.74	14.65	15	14.665
Hg Yellow I	5769.60	15.85	13	15.863
Hg Yellow II	5790.66	15.90	36	15.936

$$\lambda = \lambda_0 + C / (d - d_0)$$

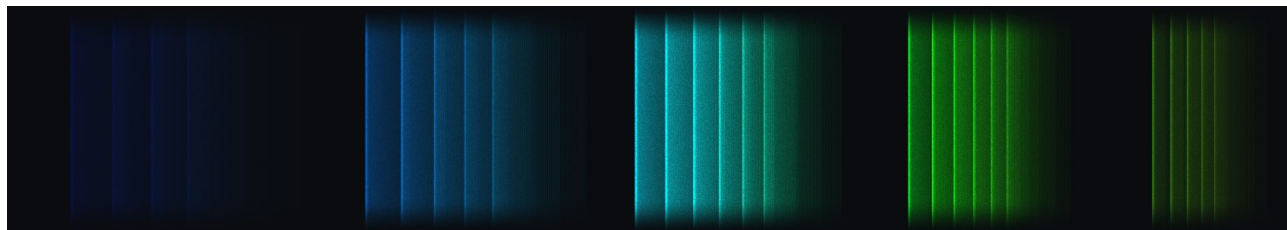
Constants worked out from three mercury lines: $\lambda_0 = 2554.66$ Å, $C = -36223.1$ Å·cm, $d_0 = 27.1298$ cm

Working: $R = (I_1 - I_2)(d_3 - d_2) / (I_2 - I_3)(d_2 - d_1) = 1.8423$, $d_0 = (R d_1 - d_3) / (R - 1) = 27.1298$ cm, then C from the first pair and I_0 from the first line.

Table 2 - the same constants applied to every mercury line measured

Line	d (cm)	λ calculated (Å)	λ standard (Å)	difference (Å)
Hg Violet	2.850	4046.56	4046.56	0.00
Hg Violet (weak)	3.350	4077.93	4077.83	0.10
Hg Blue	7.047	4358.35	4358.33	0.02

Line	d (cm)	λ calculated (Å)	λ standard (Å)	difference (Å)
Hg Blue-green (weak)	11.790	4916.04	4916.04	0.00
Hg Green	14.665	5460.69	5460.74	-0.05
Hg Yellow I	15.863	5769.69	5769.60	0.09
Hg Yellow II	15.936	5790.66	5790.66	-0.00



The AIO band system as recorded on the plate. Each mark is a band head measured with the comparator.

Table 3 - AIO band heads

Band (ν' , ν'')	$\Delta\nu$	d (cm)	λ (Å)	$\bar{\nu}$ (cm^{-1})
(2, 0)	+2	8.219	4470.13	22370.7
(3, 1)	+2	8.446	4493.40	22254.9
(4, 2)	+2	8.658	4515.65	22145.2
(5, 3)	+2	8.853	4536.58	22043.1
(1, 0)	+1	9.825	4647.90	21515.1
(2, 1)	+1	10.019	4671.63	21405.8
(3, 2)	+1	10.199	4694.14	21303.2
(4, 3)	+1	10.363	4715.07	21208.6
(5, 4)	+1	10.516	4734.96	21119.5
(0, 0)	0	11.294	4842.08	20652.3
(1, 1)	0	11.460	4866.31	20549.5
(2, 2)	0	11.612	4888.95	20454.3
(3, 3)	0	11.752	4910.20	20365.7
(4, 4)	0	11.879	4929.82	20284.7
(5, 5)	0	11.994	4947.87	20210.7
(0, 1)	-1	12.783	5079.48	19687.0
(1, 2)	-1	12.912	5102.39	19598.7
(2, 3)	-1	13.030	5123.71	19517.1
(3, 4)	-1	13.136	5143.17	19443.3
(4, 5)	-1	13.234	5161.43	19374.5
(5, 6)	-1	13.319	5177.47	19314.5
(0, 2)	-2	14.113	5337.46	18735.5
(1, 3)	-2	14.212	5358.78	18660.9
(2, 4)	-2	14.300	5378.02	18594.2

Band (v' , v'')	Δv	d (cm)	λ (Å)	$\bar{\nu}$ (cm ⁻¹)
(3, 5)	-2	14.380	5395.73	18533.2
(4, 6)	-2	14.450	5411.42	18479.4

Deslandre tables

Table 4 - band heads in Å

$v' \setminus v''$	0	1	2	3	4	5	6
0	4842.1	5079.5	5337.5	-	-	-	-
1	4647.9	4866.3	5102.4	5358.8	-	-	-
2	4470.1	4671.6	4888.9	5123.7	5378.0	-	-
3	-	4493.4	4694.1	4910.2	5143.2	5395.7	-
4	-	-	4515.6	4715.1	4929.8	5161.4	5411.4
5	-	-	-	4536.6	4735.0	4947.9	5177.5

Table 5 - band heads in cm⁻¹

$v' \setminus v''$	0	1	2	3	4	5	6
0	20652.3	19687.0	18735.5	-	-	-	-
1	21515.1	20549.5	19598.7	18660.9	-	-	-
2	22370.7	21405.8	20454.3	19517.1	18594.2	-	-
3	-	22254.9	21303.2	20365.7	19443.3	18533.2	-
4	-	-	22145.2	21208.6	20284.7	19374.5	18479.4
5	-	-	-	22043.1	21119.5	20210.7	19314.5

Upper electronic state B²Σ⁺

First differences $\Delta G'(v + \frac{1}{2})$, cm⁻¹

Interval	Value taken
$\Delta G'(0 + \frac{1}{2})$	862.8
$\Delta G'(1 + \frac{1}{2})$	855.9
$\Delta G'(2 + \frac{1}{2})$	848.9

Second difference and the constants

Quantity	Value
$\Delta^2 G' = \Delta G'(\frac{1}{2}) - \Delta G'(\frac{3}{2})$	6.90
$\omega_e x_e'$ (cm ⁻¹)	3.45
ω_e' (cm ⁻¹)	869.7
x_e'	0.00397

Lower electronic state $X^2\Sigma^+$

First differences $\Delta G''(v + \frac{1}{2})$, cm^{-1}

Interval	Value taken
$\Delta G''(0 + \frac{1}{2})$	965.3
$\Delta G''(1 + \frac{1}{2})$	951.4
$\Delta G''(2 + \frac{1}{2})$	937.3

Second difference and the constants

Quantity	Value
$\Delta^2 G'' = \Delta G''(\frac{1}{2}) - \Delta G''(\frac{3}{2})$	13.90
$\omega_e'' x_e''$ (cm^{-1})	6.95
ω_e'' (cm^{-1})	979.2
x_e''	0.00710

Result

Table 6 - vibrational constants of AIO

Quantity	My value	Literature	Difference (%)
ω_e' (cm^{-1}), upper state $B^2\Sigma^+$	869.7	870.0	0.0
$\omega_e' x_e'$ (cm^{-1})	3.45	3.50	1.4
x_e'	0.00397	0.00402	1.3
ω_e'' (cm^{-1}), lower state $X^2\Sigma^+$	979.2	979.2	0.0
$\omega_e'' x_e''$ (cm^{-1})	6.95	6.97	0.3
x_e''	0.00710	0.00712	0.3

The vibrational quantum of the excited state is smaller and its anharmonicity larger, which is what one expects: the bond is weaker and the potential well shallower once the molecule is electronically excited.

Sources of error

Setting the crosswire on a shaded head is the largest single source of error; the two yellow mercury lines are close enough that the field had to be narrowed to separate them.

Questions

1. What do you mean by an electronic band spectrum?

Answer 1 as written in the record book.

2. What do you mean by the vibrational constants of AIO?

Answer 2 as written in the record book.

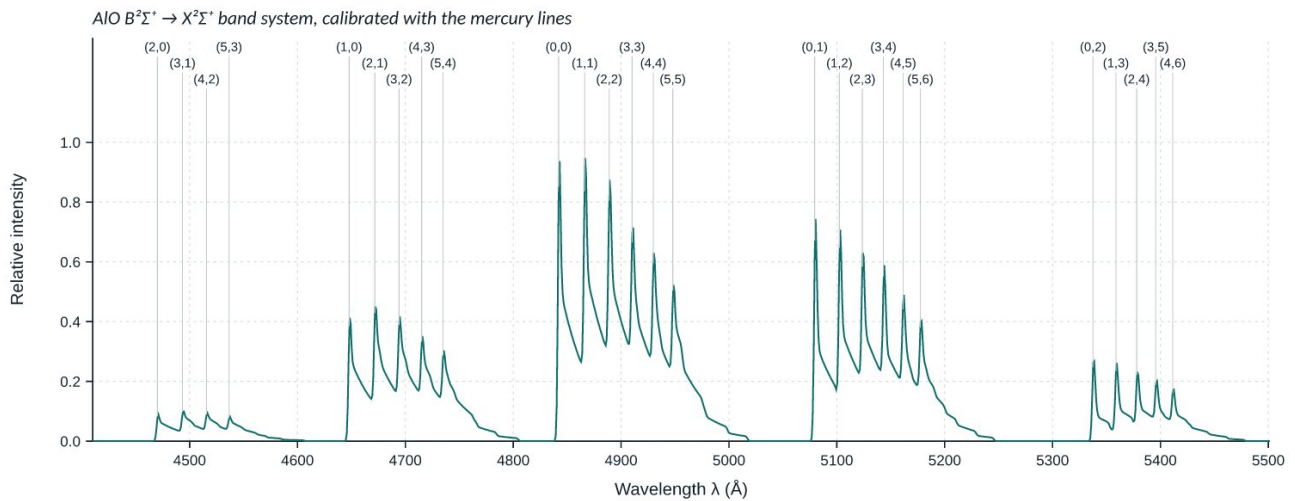
3. Do you expect a band spectrum from a homonuclear diatomic molecule?

Answer 3 as written in the record book.

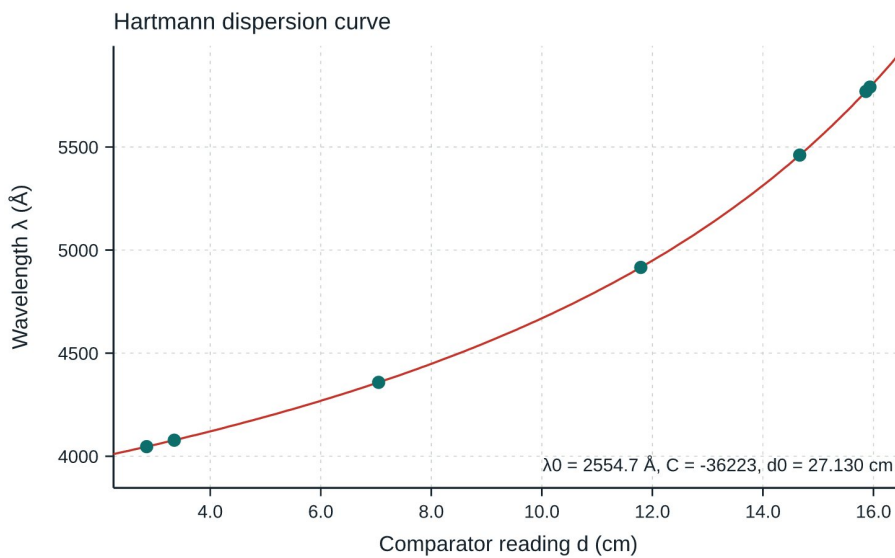
4. Explain sequences and progressions.

Answer 4 as written in the record book.

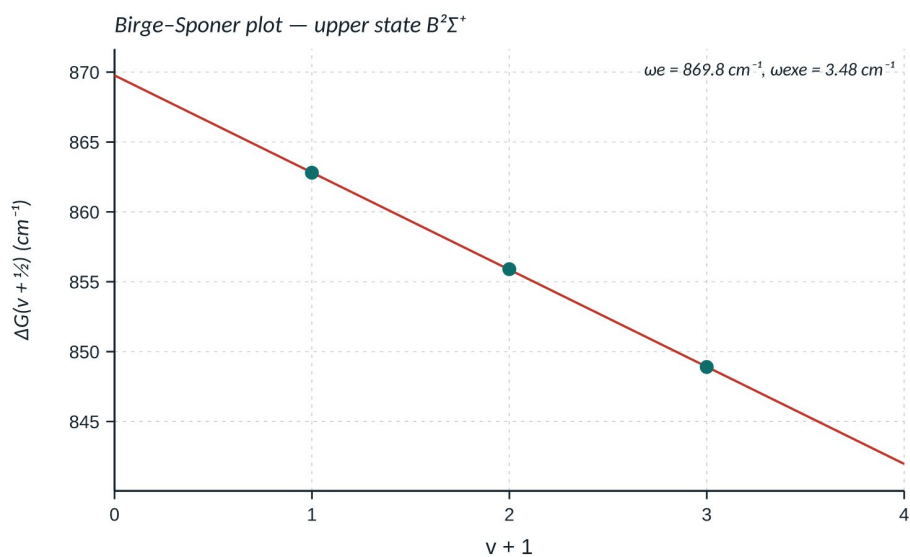
Graphs



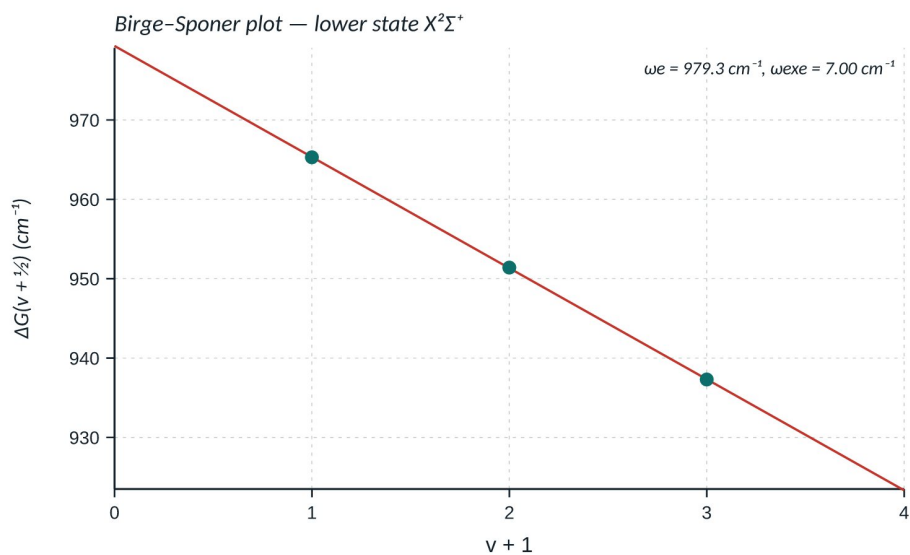
The densitometer trace of the same plate, on a wavelength scale from your own calibration.



Hartmann dispersion curve through the mercury lines.



Birge-Sponer plot for the upper state.



Birge-Sponer plot for the lower state.